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Problem Chosen
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2023 HiMCM
Summary Sheet

Electrifying a Metro Bus Fleet Without Reducing Service

Question and baseline. We study conversion of TransLink's fixed-route bus service to fully electric operation by 2033, with every 2023 departure, peak passenger capacity and accessible service preserved. Existing trolleybuses already count as electric. The comparator is ordinary renewal of the same service, not continued use of worn-out buses.

Model. A multi-year fleet replacement model links purchases and retirements to exact trip and charging feasibility. Each candidate must satisfy vehicle-class compatibility, departure times, combined passenger capacity, winter battery reserve, deadhead travel, spare buses and depot electrical limits. Gross discounted cost is kept separate from agency payment after 0%, 25% or 50% grants. Environmental scoring uses matched life-cycle greenhouse-gas boundaries. Only feasible candidates belong on the cost–emissions Pareto frontier.

Observed result. Official route data contain 191 routes and 5,466,190.5 hours, covering 98.035% of independently reported 2023 fixed-route bus hours. Actual service totals are 106,868,324 km and 5,715,791,401 capacity km after SeaBus is excluded. December reporting identifies 280 trolley/BEB buses combined; exact vehicle ages, blocks and site power remain unresolved. City-wide 2033 feasibility and Pareto scores remain *unverified*.[12, 6]

Critical technology and planning finding. A historical TransLink battery-bus scenario uses 0.02 L/km of fuel for cabin heat and fails the problem's strict fully electric definition. At 68,000 km/year, that is 1,360 L of heater fuel. Later named budgets also plan 203 CNG/diesel replacement vehicles across distinct cohorts; if delivered, those vehicles require accelerated withdrawal for a strict 2033 endpoint.[8, 12, 13]

Solved planning result. London's registration cohorts yield a numerical ten-year benchmark: retaining the original available fleet by body type, a 17-year chassis-life scenario needs annual renewal capacity of at least 776 original vehicle positions; a 14-year scenario needs 977. Both integer-program optima equal an independently derived deadline bound. Route and charging checks determine additional vehicles. Separately, 243 charging LP cases show how grid capacity changes tariff choice: a conditional 100-distance-equivalent case needs at least 4.140 MW for an eight-hour window; its overnight electricity bill is 32.96% below general service, before capital.

Recommendation. Treat 2033 as an accelerated target to test against TransLink's published 2040 goal. Obtain the missing block and depot data, retain trolley service, convert proven standard-bus blocks first, and require all-electric winter trials before committing double-deck and shuttle blocks.[8, 15]

Keywords: electric bus; fleet replacement; service preservation; charging; life-cycle cost.

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1 Problem definition and study boundary

The competition asks for environmental and financial consequences, a ten-year transition in three metropolitan areas, and a decision-maker letter.[1] We take 31 December 2023 as Vancouver's decision baseline and plan purchases and construction in 2024–2033. Documents published in 2024–2026 inform later sensitivity checks, but their realized purchases are not backdated into 2023 decisions. Metro Vancouver is the primary case; its census metropolitan area had 2,642,825 residents in 2021, exceeding the problem's 500,000-person threshold.[4] Chicago and London test transferability using their separately dated 2023 fleet snapshots; London's March audit is not treated as December stock.

The problem's definition requires propulsion *and accessories* to use a zero-emission electric source. Thus a battery bus with a diesel auxiliary heater is not fully electric in this study. TransLink's historical depot-BEB cost table assumes precisely such a heater at 0.02 L/km; its 1.4 kWh/km electricity figure cannot be carried over unaltered to an all-electric winter bus. Renewable diesel, renewable natural gas, hybrid and hydrogen fuel-cell buses may reduce emissions, but do not satisfy this strict endpoint. Already electric trolleybuses do, provided their operations remain feasible. TransLink itself has a 2040 zero-emission target, making the requested 2033 date an accelerated counterfactual.[8, 15]

The decision unit is year y, bus class k, depot d, trip i, vehicle v and charger interval t. Classes include standard, articulated, highway coach, double-deck, community shuttle and trolley vehicles. We preserve 2023 fixed-route coverage and do not offset electrification cost by reducing service.

2 Evidence and baseline audit

2023 observation	Value	Source/status
Published route identifiers	191	official CSV[17]
Route service hours	5,466,190.5	official CSV[17]
Actual fixed-route bus hours, excluding SeaBus	5,575,772	2023 actual[12]
Actual fixed-route bus service km	106,868,324	same source[12]
Actual fixed-route bus capacity km	5,715,791,401	same source[12]
Route-file share of exact bus hours	98.035%	calculated
Route/time-bin frequency rows; nonmissing	6,261; 6,041	official CSV[21]
Route/direction/time-bin load rows; nonmissing	11,659; 9,371	official CSV[22]
Revenue-fleet operating GHG, all modes	112,591 t CO_2e	2023 report[23]
CNG fleet count	299	2023 report[23]
Trolley plus battery-electric buses	280	Dec 2023 notice[6]
Battery-electric buses	about 5	Dec 2023[7]
Trolley buses	262	Feb 2023 notice[5]

Table 1: Public service data support the order of magnitude, not an exact timetable.

The 2025 budget's 2023 actual service table includes SeaBus in its published transit subtotal. We subtract 13,713 SeaBus hours, 185,370 km and 71,367,323 capacity km to obtain the fixed-route bus values in Table 1; the source's five operating-group rows reconcile exactly in the accompany-

ing audit. The implied service-km-weighted average capacity is 5,715,791,401/106,868,324 = 53.48 passenger spaces per bus-km, an aggregate calibration target rather than a standard-bus specification. The earlier statutory report rounds bus hours to 5,577,000, yielding a slightly different 98.013% route coverage check. The route data exclude discontinued routes. Time bins are coded periods, not one-hour records, and reported peak load is an average of observed peaks rather than the maximum on every departure. Route revenue hours times average speed yield a 93.74-million-km *estimator*, 87.716% of the independently reported actual bus service km. The difference cannot be attributed wholly to deadhead without vehicle-block data.[12, 20]

The 2018 official service guidelines define peak load factor as average peak passengers divided by practical vehicle capacity. Their reference capacities are 24 (mini-bus), 47 (standard trolley), 50 (standard bus or coach), 70 (articulated trolley), and 75 (articulated bus). Dividing each positive 2023 time-bin load by its load factor reproduces one of these values in 8,705 of 9,371 cells; a further 344 cells give exactly 85, consistent with TransLink's separately described 80–86-seat double-deck vehicles. Thus 9,049 cells (96.56%) match six capacity clusters, and 322 do not. Across 180 routes with positive cells, 34 show more than one cluster or an unmatched cell. This is a useful *aggregate vehicle-class diagnostic*, not a trip-level vehicle assignment: capacities are historical averages, the 85-person class is inferred, and mixed or differently averaged cells cannot establish an individual bus's capacity. The reproducible cell-level calculation and every unmatched value are retained in the data supplement.[18, 19, 22]

The 2023 accountability report also records 112,591 tonnes CO_2e of *revenue-fleet operating* emissions, comprising 110,402 Scope-1 and 2,189 Scope-2 tonnes; it separately reports 33,718 tonnes of biogenic CO_2 from renewable fuels. The revenue-fleet number covers bus and non-bus modes, excludes vehicle and station manufacture, and reflects that year's renewable-fuel mix (96% of purchased bus natural gas was renewable). It is therefore an independent accounting check and an upper bound on the included bus operating inventory, not a substitute for the 2024–2033 same-service, life-cycle normal-renewal baseline.[23]

The 2020 TransLink fleet transition plan records 1,672 CMBC vehicles, including 188 standard and 74 articulated trolleybuses.[8] These are historical values, not a verified 2023 fleet register. The February 2023 notice gives 262 trolleys, while a December official article says approximately 275 trolleys and five BEBs. A same-day release gives only their combined count, 280. Subtracting February's 262 from December's 280 would imply 18 BEBs, but this crosses reporting dates and conflicts with the explicit December article. We therefore use no exact December BEB stock count.[5, 6, 7] A June 2023 Board report lists 2023–24 procurement projects for 46 articulated hybrids, 15 standard BEBs, 50 CNG buses, 168 community shuttles and 25 double-deck buses; it records the 25 double-deck buses as accepted on a Q2 2023 timeline, while the other group totals were mostly outstanding. These are procurement states, not a year-end fleet roster.[10] A valid 2033 purchasing table still requires an age, class and depot register.

A separate June 2023 TransLink presentation places four BEBs and 262 trolleys in service *at that date*. It gives screening prices of approximately CAD 1.2 million per BEB and CAD 0.46–0.61 million per on-route charger. Its then-current capital concepts put Hamilton Route 100 charging at CAD 2.3 million, Port Coquitlam plus ten on-route chargers at CAD 46 million, and a Marpole depot sized for 340 BEBs at CAD 500 million. A newly built 500–600-bus-equivalent depot was screened at CAD 1.2–1.8 billion including land and development. These are 2023 planning envelopes with different scopes, *not* delivered item prices or additive line items; their commissioning dates changed in later plans. They supply bounded infrastructure cases while

site-specific electrical capacity and construction cash flow remain missing.[11, 15]

A 2025 capital-program audit narrows the age uncertainty: 188 standard trolleys entered service in 2006–07 and 73 articulated trolleys were expected to retire in 2028–29. It identifies projects tied to 57 diesel-hybrids and 50 CNG buses reaching end of life in 2023, 126 diesel-hybrids in 2024, 36 diesel-hybrids and nine highway coaches in 2025, and 39 articulated diesels in 2026. The plan specifies 57 BEBs for 57 retiring hybrids, seven BEBs for six retiring hybrids, and 45 articulated BEBs for 39 retiring diesels in three separate programs. These are retrospective planned cohorts, not confirmed 2023 stock, deliveries, or a route-feasible 2033 solution. The 57-BEB project's CAD 106.846 million gross capital budget includes CAD 102.498 million of reported external funding; divided by 57, its *gross project budget* is CAD 1.874 million per planned bus. The other identified BEB projects yield CAD 1.867 million per planned standard bus and CAD 3.037 million per planned articulated bus on the same budget/count arithmetic. These are project envelopes, not delivered bus-only unit quotes. The accompanying cohort audit reproduces the ratios and divisions from the published table.[12]

As a bounded funding illustration using only the 57-BEB project's gross budget, assuming *all* CAD 106.846 million is grant-eligible, 0%, 25% and 50% hypothetical grants leave agency payments of CAD 106.846, 80.135 and 53.423 million, respectively. The gross project budget remains CAD 106.846 million in each case. These are undiscounted funding splits of one later-reported program, not incremental economic scores: eligibility, spending dates, and the same-service conventional renewal budget are unresolved. The actual CAD 102.498 million external-funding entry is a distinct retrospective budget fact and does not override the problem's 50% scenario limit.[12]

A subsequent 2026 plan changes the retrospective picture further. A named project now proposes 39 60-foot diesel buses, 64 in-route-charged 40-foot BEBs and 153 depot-charged 40-foot BEBs for 236 retiring diesels: 256 planned replacement vehicles, or 1.085 per retirement. Another cohort proposes 31 BEBs for 27 retiring hybrids, a 1.148 ratio. The 2025 plan's 45 articulated BEBs for 39 retiring diesels is an earlier description with possible cohort overlap; its count is *not* added to the later project. The 2025 plan separately names 50+84+30=164 CNG replacements for 2023–2025 retirement cohorts. Thus these distinct named programs plan $164 + 39 = 203$ non-electric replacement vehicles associated with 2023–2026 retirement cohorts. If procured and still operating, all 203 would need accelerated removal to achieve the strict 2033 endpoint. This is a planning-exposure count, not a verified delivery count or a complete 2033 fleet requirement.[12, 13]

To show the size of the early-retirement term without inventing a 2026 bid, suppose only the 39 planned 60-foot diesels entered service at the start of 2026 and were withdrawn at the start of 2033. The historical TransLink assumptions are a 17-year life and an approximately CAD 1.0-million 60-foot diesel price in 2019 currency. A straight-line unused-capital proxy would be $39(1.0)(10/17) = 22.94$ million in *2019 CAD before resale*. This is not an observed loss or an incremental project cost: resale/redeployment, actual purchase prices and the choice to avoid these purchases in a 2023 decision change the economic comparison. It shows why premature retirement cannot simply be omitted.[8, 13]

3 Fleet and service model

3.1 Decision variables and normal-renewal baseline

Let x_{ykd} be electric buses purchased; r_{ykda} conventional buses retired from age cohort a; h_{ydq} charger positions of power type q commissioned; and z_{iv} equal one when bus v performs baseline trip i. Let c_{vt} denote charge energy and e_{vt} battery energy. For the normal-renewal case, age-expired conventional buses are replaced with conventional equivalents while the *same trip and passenger-capacity demand* is served. An electric candidate is evaluated only against this feasible comparator. Historical purchase expenditure of existing buses is sunk; early retirement enters through foregone residual service or sale value.

For each class and depot, stock obeys

$$N^p_{y,k,d} = N^p_{y-1,k,d} + x^p_{y,k,d} - r^p_{y,k,d}, \qquad p \in \{\text{electric}, \text{conventional}, \text{trolley}\}. \tag{1}$$

No diesel, hybrid, CNG or hydrogen propulsion or non-electric accessory may operate on any fixed-route trip in 2033. Retirement and purchase are not constrained one-for-one: needed vehicle counts emerge from schedules and spare ratios. A short-range vehicle may force additional buses, chargers or drivers.

3.2 Hard service and technical constraints

Every baseline trip is retained at its original route, direction and departure time. Multiple smaller buses may serve one departure when necessary, and each assigned class must be compatible with the route:

$$\sum_{v \in V_i} z_{iv} \geq 1 \qquad (i \in I), \tag{2}$$

$$\sum_{v \in V_i} Q_v z_{iv} \geq Q^0_i \qquad (i \in I), \tag{3}$$

$$\sum_{i \in I_{r,s,t}} \sum_{v \in V_i} Q_v z_{iv} \geq Q^0_{r,s,t}. \tag{4}$$

The final expression denotes capacity available at the original route–direction peak; the trip-level inequality protects each departure. A smaller bus can replace a larger one only if an additional vehicle on that departure restores the capacity and both vehicles pass scheduling, energy and site checks. Accessible trips require accessible vehicles; depot/class spare stocks are protected. If demand growth is modeled, it is applied equally to baseline and electric cases.

As a separate *calibration check*, the assigned annual output should reproduce 5,575,772 fixed-route bus service hours, 106,868,324 service km and 5,715,791,401 capacity km for the 2023 reference service within documented differences in measurement definitions. The capacity-km subtotal is especially useful for detecting systematic undersupply across small-bus substitutions, but it cannot substitute for the departure and peak constraints above.[12]

Each vehicle's assigned trips and charging intervals cannot overlap. Location-to-location deadhead times must fit between events, including departure from and return to its depot. Driving to or from service also removes a vehicle from the available depot spare count. For battery vehicles, energy follows

$$e_{v,t+1} = e_{vt} + \eta_q c_{vt} - \epsilon_{vk}^{\text{traction,winter}}(\ell_{it} + \ell_{vt}^{\text{deadhead}}) - E_{vt}^{\text{accessories}}, \tag{5}$$

$$\underline{s}_v B_v \le e_{vt} \le \bar{s}_v B_v. \tag{6}$$

Here $E_{vt}^{\text{accessories}}$ includes all-electric cabin heating and other onboard equipment during trips, deadhead, charging and active layovers; the traction coefficient excludes that energy to prevent double counting. An unmeasured winter accessory load makes an electric schedule `UNVERIFIED`, not zero-energy. Battery capacity and passenger capacity are linked when battery weight reduces available seats or payload. Charger occupancy, power and opening year impose

$$\sum_v u_{vdt} \le H_{dt}, \quad \sum_v P_{vdt} \le P_{dt}^{\text{grid}}, \quad H_{dt} = 0 \text{ until commissioned.} \tag{7}$$

Conventional baseline vehicles instead require a fuel budget, with jointly calibrated distance and elapsed-duty rates a_v, b_v in a consistent fuel unit (L or kg):

$$F_{v,t+1} = F_{vt} - a_v \ell_{vt} - b_v \Delta t, \qquad F_v^{\text{reserve}} \le F_{vt} \le F_v^{\text{tank}}. \tag{8}$$

The implemented check assumes no refuelling within a duty; the time rate includes stationary consumption and must not duplicate the distance rate. Missing fuel data leave the normal-renewal comparator unverified. Such vehicles may serve intermediate years but cannot satisfy the strict 2033 endpoint.

Trolleybuses use a separate wired-network check. Each trip, deadhead and powered layover needs a certified connected path and an operating feeder. Concurrent demand must obey $\sum_{v \in V_{ft}} P_v \le P_f^{\text{firm}}$, where firm power is available after other loads and reserve. The code conservatively reserves each vehicle's measured all-in peak draw on every feeder traversed for the entire moving segment; stationary accessories use their separate demand. A failure of this sufficient power certificate calls for finer feeder timing before declaring physical infeasibility. Missing path or feeder records leave existing trolley operation unverified, without assigning these vehicles fictitious batteries or buying their capacity again.

Depot bus parking capacity, charger footprint, charging efficiency, construction lead time and route-specific winter testing must also pass. A charger site may be the home depot or a route terminal, provided the location and travel-time checks pass. The accompanying validator checks the supplied trip, energy and site records, returning `UNVERIFIED` for missing essential inputs; it cannot certify a city-wide solution from time-bin averages.

3.3 Optimization order

For each scenario, first solve or test service feasibility. Second find the minimum gross-cost normal-renewal baseline. Third enumerate feasible electric strategies: depot charging, terminal fast charging, battery sizing, depot sequencing and additional buses. Fourth solve an ϵ-constraint family $\min C$ subject to $G \le \epsilon$ to reveal the Pareto frontier. A displayed "balanced" option would be chosen only after that frontier and stakeholder weights are available. We do not present a fictional optimum from incomplete inputs. This joint approach follows the scheduling/fleet-size logic of Rogge et al., the multi-year replacement logic of Pelletier et al., and the multi-depot timing logic of Gkiotsalitis et al.[24, 25, 26]

4 Costs, funding and environmental accounting

Let K_y include bus and battery procurement (initial battery included in a quoted vehicle price), chargers, depot civil work, grid connection and distribution upgrades, construction/design, training and any extra buses. Let O_y include electricity, fuel, maintenance, replacement batteries, drivers, demand charges and regular charging-site upkeep. Let S_y be realized salvage and R_{2033} the remaining value of vehicles and durable infrastructure at the horizon. With a stated real discount rate r and consistent price year,

$$C^{\text{gross}} = \sum_{y=2024}^{2033} \frac{K_y + O_y - S_y}{(1+r)^{y-2023}} - \frac{R_{2033}}{(1+r)^{10}}, \tag{9}$$

$$\Delta C = C^{\text{gross}}_{\text{electric}} - C^{\text{gross}}_{\text{renewal}}. \tag{10}$$

For grant rate $f \in \{0, 0.25, 0.50\}$ and eligible expenditures E_y, $C^{\text{agency}} = C^{\text{gross}} - \sum_y fE_y/(1+r)^{y-2023}$. A grant changes who pays, not gross buses, copper, electricity or labour. Local employment and procurement impacts require an additional regional economic model and are reported separately to avoid double counting.[27]

Define a life-cycle greenhouse-gas inventory over equal service:

$$G = G_{\text{vehicle+battery}} + G_{\text{infrastructure}} \tag{11}$$

$$+ \sum_y \left(G_{\text{fuel,y}} + G_{\text{electricity,y}} + G_{\text{maintenance,y}}\right) - G_{\text{end of life}}. \tag{12}$$

Electricity uses an explicitly selected grid-intensity scenario. British Columbia's 2023 integrated-grid factor is 0.0113 kg CO_2e/kWh; it is an annual average and should be varied against marginal-generation possibilities.[28] On-route NO_x, particles, electricity use and battery-production impacts appear in separate columns rather than being concealed by a single carbon score. If compatible manufacturing/infrastructure factors are absent, the *life-cycle* score is unavailable even when operating emissions can be estimated.

The requested scores are

$$S_C = 100\frac{C_{\text{renewal}} - C_{\text{electric}}}{C_{\text{renewal}}}, \qquad S_G = 100\frac{G_{\text{renewal}} - G_{\text{electric}}}{G_{\text{renewal}}}. \tag{13}$$

Negative values remain negative. The economic score is based on *gross* cost, with subsidized agency cost shown beside it; otherwise the 50% grant would artificially change the societal merit ranking.

5 Observed and illustrative results

The route table covers 98.035% of independently reported actual fixed-route bus hours, and the source supplies actual annual bus service and capacity km. This passes stronger scale checks but not departure, block, winter energy or concurrent-charging checks, which require a missing vehicle-duty manifest. The 2023 full-city feasibility state is `UNVERIFIED`; no verified full-city score, purchasing schedule or Pareto set is issued.

For a transparent *historical parameter demonstration*, the 2020 plan lists a 2019-CAD 40-foot diesel purchase of CAD 630,000, a depot BEB purchase of CAD 1,110,000, a 50-kW depot charger

position of CAD 298,000, a 0.60 L/km diesel rate, a 1.4 kWh/km BEB electricity rate, 0.02 L/km of BEB heater fuel and 68,000 km per bus annually.[8] One-to-one vehicle purchase premium is CAD 480,000; 68,000 km implies 40,800 L for diesel, versus 95,200 kWh *plus 1,360 L heater fuel* for the historical BEB. At BC's 2023 *average* factor, its electricity alone corresponds to about 1.076 t CO_2e/year. The BEB's heater makes it ineligible for the 2033 endpoint, and replacing that heater with electric heat changes winter energy demand. The CAD 298,000 charger quotation already includes design, permitting, project management, financing and a construction contingency; those items must not be charged again. This is neither a 2023 bid nor a complete cost/emissions comparison. TransLink's historical planning replacement ratio of 1.15 for depot charging shows why one-to-one replacement is unsafe.[8]

For a separate *electric-only 40-foot reference case*, the same plan models in-route charging at CAD 4.86/km, versus CAD 4.62/km for diesel, 4.77/km for hybrid and 4.15/km for CNG, all in 2019-CAD lifetime cost. The three alternative economic scores are −5.19%, −1.89% and −17.11% respectively. The December 2020 update explicitly uses hybrid replacements as its conventional base case for retiring diesel/hybrid buses and CNG replacements for retiring CNG buses; the diesel comparison is a sensitivity reference. None is a fleet score: normal-renewal vehicle type, route assignment and replacement timing determine the actual denominator. Its in-route table specifies 1.6 kWh/km with no auxiliary fuel; using 0.60 L/km for diesel, the federal heavy-duty diesel combustion factor of 2.630 kg CO_2e/L and BC's 2023 average grid factor of 0.0113 kg CO_2e/kWh gives an *operational-only* diesel-to-BEB GHG score of 98.85% at identical distance. At 68,000 km/year, that is 106.07 t CO_2e/year of modeled operating savings per equivalent bus. Charging-site availability and actual 2023 blocks remain unverified; vehicle/battery/infrastructure manufacturing and fuel supply are excluded, so the requested full life-cycle city-wide environment score remains unavailable.[8, 9, 29, 28]

The 2024 ZEFTP offers a separate retrospective check. Its operating-and-maintenance costs are 3.30, 1.97, 2.77 and 2.28 CAD/vehicle-km for diesel, hybrid, CNG and BEB, respectively. On that *O&M-only* boundary, BEB scores are +30.91%, −15.74% and +17.69% against the three alternatives; vehicle, battery and charger capital are absent. Its electric-heater BEB is reported at 19 g CO_2e/km, versus 1580 for fossil diesel, 1420 for fossil hybrid, 1350 for fossil CNG and 250 for renewable CNG. Fuel mix therefore changes the operational environmental comparator. These 2024 values are neither 2023-known procurement inputs nor full life-cycle 2024–2033 totals; they cannot be added to the older 2019-CAD costs.[15]

An independent 2024 ICCT e-bus life-cycle method supplies transferable *sensitivity factors*: 6.6 kg CO_2e per kg of vehicle glider/powertrain, 58 kg CO_2e per kWh of LFP battery, and 52.4/67.5 g CO_2e/km for diesel/BEB maintenance. Its 12–15 m example uses a 335-kWh battery and glider masses from Latin American buses, so these are not observed TransLink specifications. Its own system boundary excludes charging infrastructure and bus recycling. We retain separate terms for these missing components and do not substitute this partial method for a completed Vancouver life-cycle inventory.[16]

Facility timing is also consequential. A December 2020 program listed 677 planned BEB purchases and CAD 246.4 million of depot/charging investments, but subsequent 2024 review targeted Port Coquitlam for over 100 BEBs in 2027 and Marpole for roughly 340 in 2028. It estimated six to eight years to deliver charging at existing transit centres and recommended commissioning six to twelve months before bus delivery. These are planning and later-review quantities, not 2023 installed charger counts. The 2024 screening capital envelope includes service expansion and has a

stated −50% to +100% accuracy range, so assigning it wholesale to our fixed-2023-service scenario would misstate incremental cost.[9, 15]

The 2020 update provides a useful *published-program arithmetic check*. Its Table 1 awards 449 depot-charged standard, 136 in-route-charged standard and 92 depot-charged articulated BEBs during 2020–2029, totaling 677 *planned awards*, not deliveries. The reported incremental vehicle-purchase premium is CAD 236.9 million nominal; Table 3 adds CAD 246.4 million of depot/charging investment; and Table 5 projects CAD 96.3 million of cumulative operating savings through 2030 relative to its hybrid-renewal baseline. The resulting undiscounted component sum is CAD $236.9 + 246.4 - 96.3 = 387.0$ million. Annual operating cells rounded to CAD 0.1 million sum to CAD 96.1 million in savings, differing by CAD 0.2 million from the printed total; we use the printed total and retain the discrepancy in the audit. The 15 awards in 2020 were funded in an earlier program. The update anticipated electric purchases for retiring diesel/hybrid buses, yet the later 2025 plan lists 84 CNG buses for a 2024 hybrid retirement cohort; this is direct evidence that the older purchase pathway was revised. These source-period amounts cannot be rebased into 2024–2033 cost, grant or GHG scores: facility spending dates, later procurement revisions, 2023 fleet stock and all-electric winter duty feasibility are absent.[9, 12]

The historical depot analysis supplies a direct *winter feasibility stress test*. With 660-kWh articulated batteries and fuel heaters, it projected depot-dependent replacement ratios of 1.13–1.55 and warned that electric-only heat could require roughly 10% more buses. Applied mechanically to the later capital project's 39 retiring articulated diesels, $\lceil 39(1.10)r_d \rceil$ gives 49–67 BEBs across hypothetical depot allocations, compared with its 45 planned depot-charged BEBs. The 4–22 difference is a *historical-technology sensitivity*, not a measured shortage: the 39 actual depot assignments, 2026 energy specifications and block schedules are unknown. It identifies a concrete winter trial and procurement question before certifying the strict 2033 endpoint.[8, 12]

5.1 Charging tariffs and the value of firm power

The original 2023 budget assumes diesel at CAD 1.66/L; this is a planning price, not an actual invoice. Its 32.17-million-L operational volume includes a broader system boundary and is not treated as fixed-route bus consumption.[2] BC Hydro's April-2023 tariff supplies energy, monthly demand and daily basic charges. RS1640 bills peak demand only during 06:00–22:00; RS1650's zero demand fee ends in March 2026 under that historical schedule. We therefore distinguish billed peak from physical grid power and do not extrapolate the temporary concession through 2033.[3]

For daily grid energy E, available charging hours T, site limit P, and rate-specific billed hours T_j, solve

$$\min M_j : \quad \sum_{t \in T} p_t \Delta t = E, \quad 0 \le p_t \le P, \quad p_t \le M_j \ (t \in T_j). \tag{14}$$

Power is zero outside T. If h_0 available hours are outside the billing window and $h_1 > 0$ are inside, the sharp lower bound is $M_j \ge \max\{0, (E - h_0 P)/h_1\}$. Physical feasibility also requires $E \le (h_0 + h_1)P$; a wholly nonbilled feasible window gives $M_j = 0$. The executed CPU LP matches this independent bound in every feasible case. A second stage minimizes physical peak at the optimal billed peak. Monthly bills include energy, basic and demand charges, the prior-winter eleven-period minimum-bill floor, and the 1% rider refund; taxes and capital are excluded.

We normalize to 100 *distance-equivalent units*, each at the historical 68,000 km/year reference. These are not an observed fleet or a one-for-one replacement count. Historical onboard energy

of 1.37–2.0 kWh/km, assumed charging efficiencies of 85/90/95%, three common return windows and 2.5/3.5/5-MW site limits give 243 scenarios. Annual energy is spread evenly across 365 days; this relaxation does not certify winter energy, battery capacity, charger-port occupancy or actual bus availability. A meter-measured intensity would not be divided by efficiency again.[8]

Rate	Physical MW	Billed MW	Annual CAD million
1600	4.140	4.140	1.350
1640	4.140	0.000	0.905
1650 (pre-Apr 2026)	4.140	4.140	1.113

Table 2: Conditional charging bounds for 100 distance-equivalent units, 68,000 km/year each, 1.6 battery-kWh/km, 90% charging efficiency, 22:00–06:00 availability and 5-MW site power. April-2023 prices are frozen for a 365-day comparison; bills include the 1% rider refund and exclude taxes and capital.

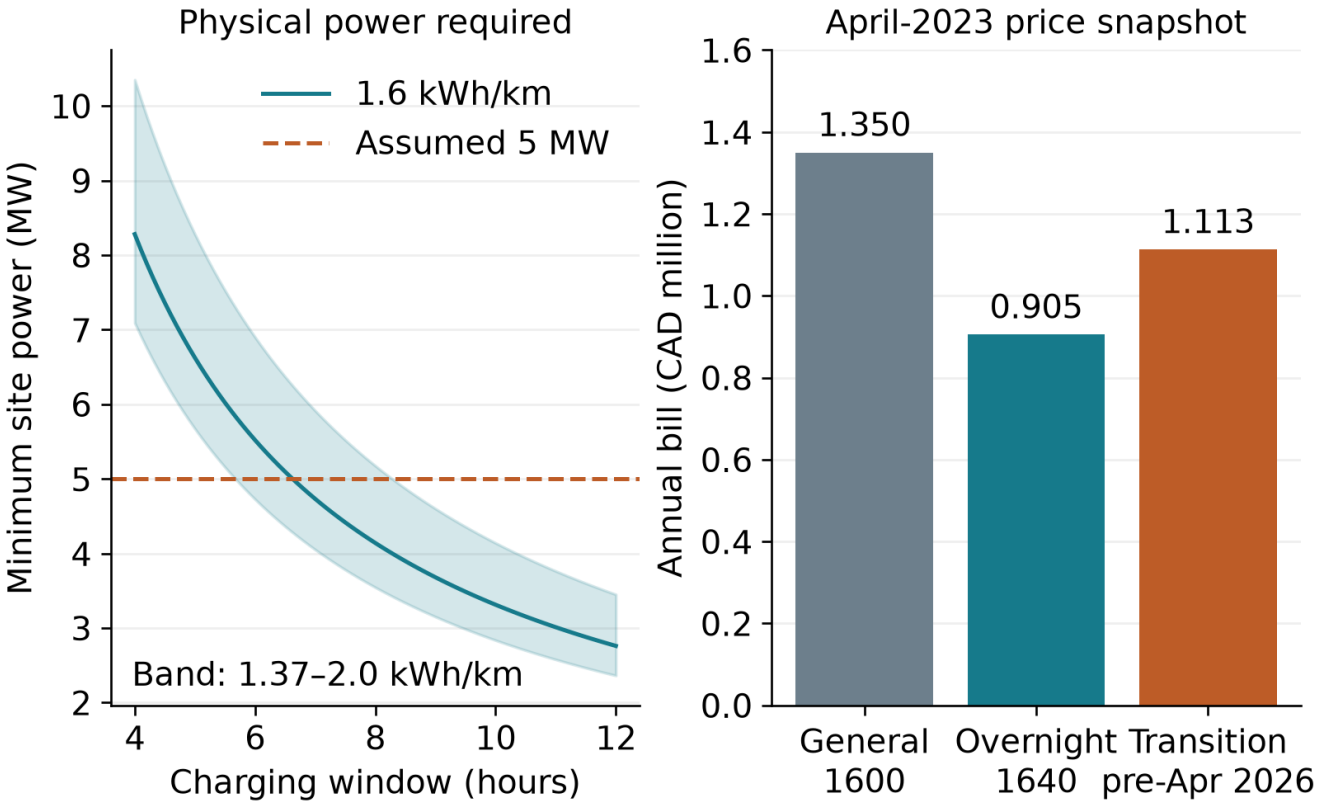


Figure 1: Physical charging-power bounds and conditional tariff bills. The left panel uses 90% charging efficiency; the right uses the central eight-hour case. A zero billed peak still requires 4.140 MW of physical power.

In the central case, an eight-hour window needs at least 4.140 MW, and RS1640 saves 32.96% of the RS1600 electricity bill. At 3.5 MW, eight-hour-only charging is impossible under the declared energy requirement. Extending availability to 20:00–06:00 restores aggregate feasibility, but RS1640 then costs CAD 1.285 million/year versus CAD 1.227 million under RS1600. The bills cross at 3.598 MW; two separate LP solutions confirm this threshold. Between 3.312 and

4.140 MW, each extra firm kW saves CAD 594/year under RS1640 in this fixed-price, fixed-window case, before paying for the upgrade. This quantifies an infrastructure contribution through feasible service and avoided bills, without crediting construction expenditure as a second benefit. Individual vehicle certification and dated future tariffs remain necessary before using these bounds in a purchasing decision.

6 Conditional 2024–2033 roadmap and transfer

Year	Decision gate and work package	Release condition
2024	Freeze 2023 fleet/class/age/depot register and exact weekday, weekend and winter trip blocks; measure spare and peak capacity.	Audited service baseline.
2025	Winter test all-electric standard and articulated buses, including cabin heat and accessible capacity. Survey depots and utility interconnections.	Tested energy bounds.
2026	Procure standard-bus replacement cohorts only on validated blocks using commissioned charging; prepare Port Coquitlam commissioning.	No missed trip or SOC breach.
2027	If commissioned, expand Port Coquitlam standard-bus blocks and test terminal charging where depot dwell is insufficient.	Site power and timetable proof.
2028	If commissioned, allocate Marpole electric blocks; complete articulated trials and publish cost recalibration.	Capacity and winter proof.
2029	Convert eligible articulated and coach blocks; protect trolley operation and renew aged trolley assets as needed.	No loss of trolley service.
2030	Decide double-deck vehicle specification and depot adaptations based on actual payload/range tests.	Full-day block proof.
2031	Commission double-deck facilities and candidate shuttle charging; revise Pareto frontier with observed tariffs and degradation.	Manufacturer and utility readiness.
2032	Convert remaining technically feasible classes; stress test outages, snow days and spare fleet.	Every block checked.
2033	Retire or convert every remaining non-electric fixed-route vehicle, including accessory heaters; certify service and emissions.	All constraints pass; otherwise disclose gap.

These are *stage gates*, not fabricated annual vehicle counts. A valid annual quantity is x_{ykd} from the model, conditional on the unresolved fleet and site data. The 2024 TransLink technical advice delays double-deck BEBs until after 2030 and says most shuttle blocks are not yet feasible, so the last two rows are high risk rather than guaranteed.[15]

The later approved 2025 Investment Plan's A9 table projects 4.424 million bus service hours in 2033. Against the independently reported 2023 actual fixed-route bus total of 5.576 million, this is a conditional 1.152-million-hour (20.66%) shortfall *if the two accounting boundaries match*. Its 2025–2034 bus-hour cells reconcile to the printed 50.049-million-hour total. We do not use that

later service trajectory as the ordinary-renewal comparator: our service floor remains the 2023 schedule, and the published plan's definitions still require reconciliation. This shows why an apparently inexpensive fleet transition cannot be credited for reduced operating service.[14, 12]

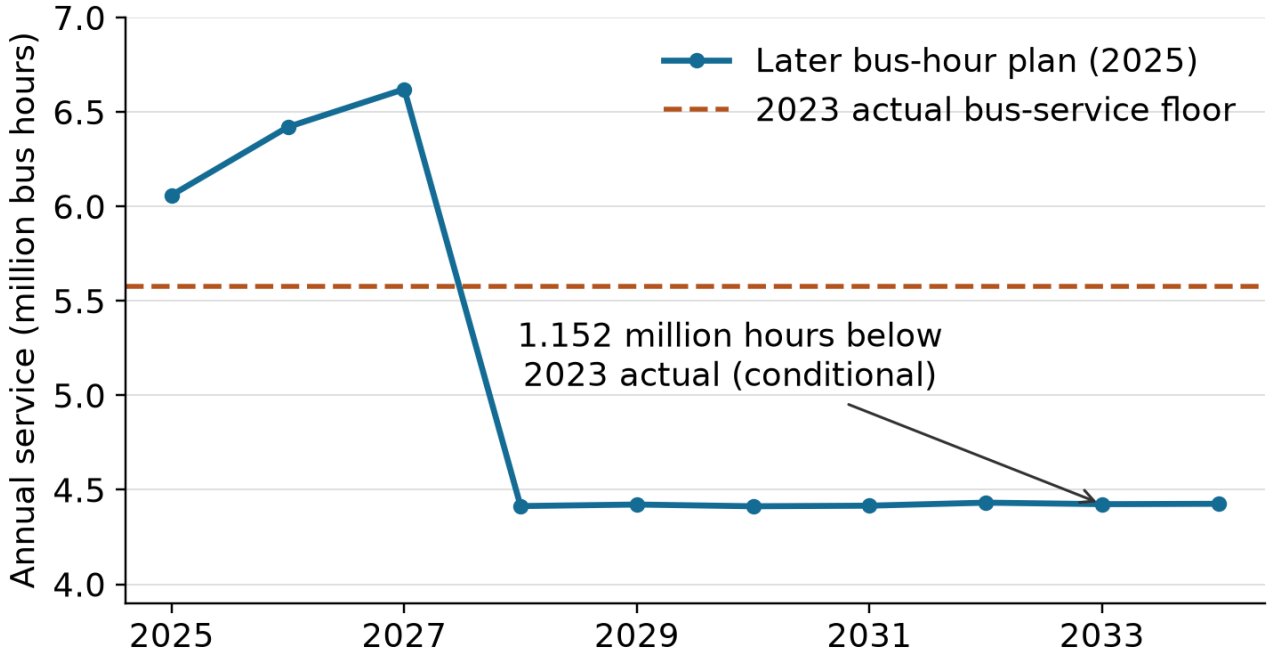


Figure 2: Conditional service-floor screen. The later plan is a retrospective comparison, not a valid fixed-2023-service electrification scenario.

Chicago's federal 2023 agency profile supplies 1,895 buses available for maximum service, 1,307 operated at maximum service, mean fleet age 11.2 years, 45,411,226 bus revenue miles (73.082 million km), and 4,970,515 bus revenue hours. The reported 45.0% spare percentage uses the operated maximum as denominator. USD 971.019 million of operating expenses covers all bus service.[30] Separately, the DOT's 2023 year-end age inventory contains 1,576 standard and 299 articulated buses, totaling 1,875; the weighted reported ages give 11.208 years. Of these, 1,056 are at least 13 years old and 539 at least 16. Under an assumed 17-year chassis-life cap and fixed availability, at least 539 original slots already require renewal in 2024; under a 14-year cap this rises to 1,056. These are scenario deadlines, not asserted CTA retirement rules. They expose the need for substantial initial capacity even before a ten-year uniform conversion is considered. Year-end inventory and maximum-service availability are distinct measures and are not silently interchanged.[31]

CTA's matched fuel table reports 13,573,058 US gallons of diesel and 479,980 kWh for battery buses. Dividing each by its own reported propulsion mileage gives 0.825 L/km and 1.733 charging kWh/km, respectively. These are annual matched-table diagnostics; its combined mileage is only 85.55% of the profile's bus revenue mileage, requiring reconciliation before extrapolation. It cannot establish winter peak energy or accessory demand.[32] The June 2023 notice's expansion from an implied 25 electric buses to 47 involves a 22-bus option valued at USD 26.2 million; receipt of the original 23-bus base order is explicit. This supplies dated stock and contract evidence, while year-end electric stock, garages, tariffs and exact duties remain unresolved.[33]

London's 31 March 2023 audit contains 8,643 contracted buses: 2,445 single-deck and 6,198 double-deck, including 950 electric and 20 hydrogen. Thus 7,693 original positions require conversion

under a fixed fleet-availability policy. The source's detailed Tables 7–8 reconcile by registration year, body and propulsion. Its Table 2's printed single-deck total of 2,447 and 2017 subtotal of 341 conflict with those detailed tables; we use the reconciled 2,445 and 339, preserving the published total 8,643. Hydrogen remains outside the strict electric endpoint.[34]

6.1 A solved age-cohort planning benchmark

For London, an original *fleet slot* denotes one available vehicle position of the same body type. Renewing a slot requires suitable electric capacity; the operational model may require additional vehicles. Let n_c be the observed size and b_c the registration year of cohort c, and choose a maximum chassis-life scenario L. These caps are assumptions for sensitivity, not asserted TfL rules. A non-electric cohort must renew by $d_c = \min(2033, \max(2024, b_c + L))$. An existing electric cohort enters the renewal model only when $b_c + L \leq 2033$; the 2013 Routemaster retains its chassis vintage despite its 2022 electric retrofit.[34]

Let integer a_{cy} allocate original slots to renewal years $2024 \leq y \leq d_c$. We first minimize the required peak annual renewal capacity M:

$$\min M : \qquad \sum_{y \leq d_c} a_{cy} = n_c, \quad \sum_{c} a_{cy} \leq M, \quad a_{cy} \in \mathbb{Z}_+. \tag{15}$$

With M fixed, a second solve minimizes $\sum_{c,y}(d_c - y)a_{cy}$, avoiding avoidable early renewal. Among ties, a third solve prioritizes earlier non-electric conversion without changing either optimum. This is an executed integer program using the public cohort counts, not an assumed uniform purchase schedule.

An independent certificate follows from $D_t = \sum_{c:d_c \leq t} n_c$:

$$M^\star = \max_{t=2024,\ldots,2033} \left\lceil \frac{D_t}{t - 2023} \right\rceil. \tag{16}$$

Necessity follows because all D_t slots must fit into $t - 2023$ annual capacities. Sufficiency follows by allocating the earliest-deadline cohorts first when every cohort may renew from 2024. Later technical release dates, class-specific production limits and charging construction can increase the requirement; the certificate concerns this cohort submodel.

Life cap	Peak slots/year	Existing EV renewals	Total slots
12	2,568	735	8,428
14	977	247	7,940
17	776	66	7,759

Table 4: Optimal London fixed-availability cohort benchmarks, 2024–2033. Life caps are scenarios. Counts are renewed original fleet slots; operational feasibility can require additional buses.

All three scenarios were solved on CPU using SciPy's HiGHS mixed-integer solver; every optimization phase reaches zero gap and the peak optima match the analytic bound. With a 17-year cap, 7,693 non-electric slots convert and 66 existing electric slots renew, requiring peak capacity 776 per year. The 14-year scenario requires 977 in its busiest year and renews 247 existing electric slots. The 12-year cap creates a 2,568-slot catch-up obligation in 2024 because much of the observed fleet is already near or above that cap. It is a stress case, not a recommended lifetime.

Year	Converted slots	EV renewals	Non-electric slots left
2024	775	0	6,918
2025	776	0	6,142
2026	776	0	5,366
2027	776	0	4,590
2028	776	0	3,814
2029	776	0	3,038
2030	775	1	2,263
2031	770	6	1,493
2032	769	7	724
2033	724	52	0

Table 5: An optimal London 17-year-cap fleet-slot roadmap. Like-body available slots are maintained; this table does not assign routes or chargers.

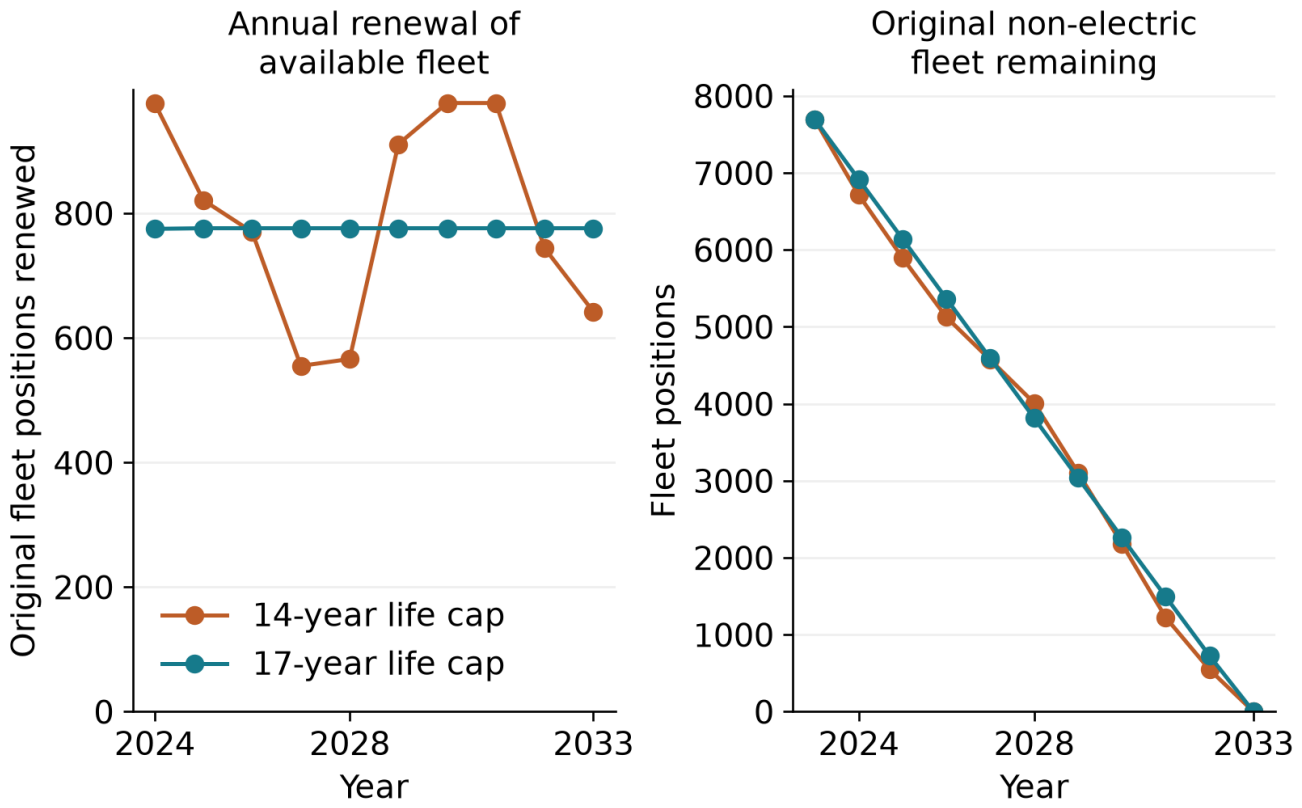


Figure 3: Executed London age-cohort roadmaps under fixed availability. The displayed numbers require compatible electric capacity, all-electric accessories, adequate chargers and trip checks before they become actual bus-purchase quantities.

The 17-year scenario advances 21,908 slot-years relative to the assumed chassis-life deadlines, compared with 334 under the 14-year scenario. These are remaining-service exposures, not monetary losses: resale and ordinary-renewal cash flows are needed to value them. The benchmark provides a numerical ten-year procurement-capacity test while retaining the operational feasibility gate.

7 Sensitivity, limitations and completion gate

The executed London life-cap experiments show why retirement policy matters: reducing the cap from 17 to 14 years raises required peak annual renewal capacity by 25.9%, while reducing the remaining-service exposure attributed to accelerated 2033 retirement. Chicago's older age profile produces a large first-year renewal condition. These age-only results do not value lost service life or certify winter operation.

For the unresolved operational baseline, solve a scenario grid over grant shares 0/25/50%, delivered bus price, energy price and demand charges, grid intensity, electric winter kWh/km, battery degradation and replacement, payload loss, charger failures, depot construction delays, natural retirement and discount rate. Vary stakeholder trade-off weights only after computing the feasible cost–GHG frontier. Report intervals and threshold values: the lowest grid capacity, highest winter energy use or longest commissioning delay at which the 2033 target fails. State the exact block, depot, lost capacity and extra vehicle/charger need for every failure.

The strongest current limitation is structural, not a lack of algebra: public TransLink time-bin service means cannot reconstruct exact timetabled trips, vehicle blocks, deadhead, energy state or charger concurrency. The 2020 fleet table cannot prove the 2023 age stock. Historical 2019-CAD cost items are not comparable with an unrebased 2023 budget. The environmental inventory requires matched production, infrastructure and electricity boundaries. Filling any of these with unsourced single numbers would turn a result into a guess.

Before a decision-ready submission, three gates must pass: (i) reconstructed service and fleet reproduce published 2023 totals within disclosed tolerances; (ii) every candidate passes trip, payload, winter SOC, charging and site constraints, including 2033 accessory power; and (iii) every cost and life-cycle emission entry has a source, year, unit and uncertainty range. The attached code already enforces the available schedule logic and returns `UNVERIFIED` when mandatory inputs are absent.

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Letter to a Metro Vancouver decision-maker

To: TransLink Board and Mayors' Council
Subject: Conditions for a service-preserving all-electric fixed-route bus fleet by 2033

Dear decision-makers,

Our analysis treats 2033 as an accelerated target relative to TransLink's published 2040 zero-emission goal. We would preserve every 2023 fixed-route departure, peak passenger capacity, accessibility and spare-bus provision throughout the transition. Existing trolleybuses already provide electric service and should be included in the baseline.

The public 2023 route data account for approximately 98.0% of independently reported actual fixed-route bus hours. We also know the 2023 total of 106.87 million bus service km and 5.72 billion capacity km. These are useful calibration targets, but they do not identify individual bus duties, fleet ages, charging opportunities or depot electrical limits. Consequently, no responsible analysis can currently certify the number of replacement buses or chargers needed to reach 2033, or quote a defensible city-wide cost and emissions score. A battery bus with a diesel heater would also fail the strict fully electric definition used here.

We recommend an audited 2023 vehicle and trip-block register, followed by all-electric winter trials. Standard buses on verified duties are the first procurement group. Articulated duties follow after capacity and winter-range checks. Double-deck and community-shuttle commitments should wait for demonstrated all-day electric operation and commissioned depot capacity. The enclosed annual roadmap makes each expansion conditional on those tests rather than relying on a one-for-one replacement assumption.

Evaluate each proposed purchase against ordinary fleet renewal with identical service. Report total project resource cost separately from TransLink's payments after 0%, 25% and 50% grants. Present both lifetime greenhouse-gas emissions and local tailpipe pollution; report battery and infrastructure manufacturing separately. Retain negative economic scores if the cleaner option costs more. A grant can improve affordability without erasing its resource cost.

Appraise electrical capacity and tariffs jointly: a zero billed nighttime peak still requires physical grid capacity. Our charging bounds show that a constrained site can reverse the preferred tariff. The decisive next step is to make the operating data available to the planning team. If winter range, vehicle payload or depot interconnection prevents any 2033 duty from being fully electric, disclose the precise gap and the additional vehicle, charger or construction condition needed. This gives the region a credible decision, even if the deadline itself proves infeasible.

Sincerely,
Fleet Transition Study Team

AI Use Report

OpenAI Codex (GPT-6) was used to locate independently published research and primary agency data, extract and audit the supplied PDFs and official CSVs, check source years and units, design the mathematical framework, implement and test the accompanying calculation code, solve the cohort scenarios and charging-power LPs, and draft and format this report.[35] The assistant did not use or search for original 2023 HiMCM solution submissions. Source values, years and limitations are recorded in the parameter ledger and source-hash manifest. The team must verify every numerical claim against the original cited material before any competition submission. No operational city-wide optimization result was generated where underlying vehicle and trip data were unavailable.